

MATH SAMPLE

FRANCIS BAHK

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1. INTEGRALS...

1.1. **Q1.** Evaluate the improper integral

$$\int_{-\infty}^{\infty} \frac{dx}{x^4 + x^2 + 1} = \frac{\pi}{\sqrt{3}}$$

Proof. We should first show that the improper integral is convergent. The integrand is continuous on the real line, so it is enough to show the integral is convergent outside a finite interval. In our case let the finite interval be $[-1, 1]$ and notice this implies for $|x| > 1$,

$$\frac{1}{x^4 + x^2 + 1} \leq \frac{1}{x^4},$$

so the integral

$$\int_1^{\infty} \frac{1}{x^4}$$

is convergent. Thus

$$\int_1^{\infty} \frac{1}{x^4 + x^2 + 1} \quad \text{and} \quad \int_{-\infty}^{-1} \frac{1}{x^4 + x^2 + 1}$$

are convergent also. Now using Proposition 5.3.2, we may write

$$\int_{-\infty}^{\infty} \frac{dx}{x^4 + x^2 + 1} = \lim_{R \rightarrow \infty} \int_{-R}^R \frac{dx}{x^4 + x^2 + 1}.$$

Our next step is to set up the contour integral. Notice following the (5.3.5) in Asmar, we have

$$I_{\gamma_R} = \int_{\gamma_R} f(z) dz = \int_{[-R, R]} f(z) dz + \int_{\sigma_R} \frac{f(z)}{dz} = I_R + I_{\sigma_R},$$

where we will use the Residue theorem to compute I_{γ_R} . Our choice of contour will be the union of the positively-oriented upper semi-circle with radius R with the path from $-R$ to R . Notice the denominator $x^4 + x^2 + 1$ has four roots defined by $x = \pm \frac{1}{2} \pm \frac{i\sqrt{3}}{2}$ according to calculation by WolframAlpha. Notice two of the four roots corresponding to $i < 0$ lie outside the contour of integration, while the other two which are the simple poles $x_1 = \frac{1}{2} + \frac{i\sqrt{3}}{2}$ and $x_2 = -\frac{1}{2} + \frac{i\sqrt{3}}{2}$, lie inside the contour. For ease of reading let $x_3 = \frac{1}{2} - \frac{i\sqrt{3}}{2}$ and $x_4 = -\frac{1}{2} - \frac{i\sqrt{3}}{2}$. Now we compute the residue of f at each simple pole. For x_1 , notice

$$\text{Res}(f, x_1) = \lim_{x \rightarrow x_1} \frac{1}{(x - x_2)(x - x_3)(x - x_4)} = \frac{1}{i\sqrt{3}(1 + i\sqrt{3})} = \frac{1}{i\sqrt{3} - 3}.$$

Multiplying by this expression by one (using the complex conjugate) yields

$$\text{Res}(f, x_1) = \frac{1}{i\sqrt{3} - 3} \left(\frac{i\sqrt{3} + 3}{i\sqrt{3} + 3} \right) = \frac{i\sqrt{3} + 3}{12}.$$

Similarly for x_2 we have

$$\text{Res}(f, x_2) = \lim_{x \rightarrow x_2} \frac{1}{(x - x_1)(x - x_3)(x - x_4)} = \frac{i\sqrt{3} - 3}{12}.$$

So with $R > 1$ the Residue theorem implies

$$\int_{\gamma_R} \frac{1}{x^4 + x^2 + 1} dx = 2\pi i \left(\frac{i\sqrt{3} + 3}{12} + \frac{i\sqrt{3} - 3}{12} \right) = \frac{\pi}{\sqrt{3}}.$$

Suppose we have x on σ_R . We will use an ML bound to show

$$\int_{\sigma_R} \frac{dx}{x^4 + x^2 + 1} \rightarrow 0$$

as $R \rightarrow \infty$. We'll notice on σ_R , $|x| = R$, so the length of the semi-circle is πR . Also notice

$$\left| \frac{1}{x^4 + x^2 + 1} \right| \leq \frac{1}{R^4 + R^2 + 1}.$$

Thus

$$\left| \int_{\sigma_R} \frac{dx}{x^4 + x^2 + 1} \right| \leq \frac{\pi R}{R^4 + R^2 + 1} = \frac{\pi}{R^3 + R + 1/R}.$$

As desired, this converges to 0 as $R \rightarrow \infty$, so $I_{\gamma_R} = \frac{\pi}{\sqrt{3}}$. $\square$

1.2. **Q2.** Evaluate the improper integral

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{e^{bx} + 1} dx = \frac{\pi}{b \sin(\frac{\pi a}{b})} \quad (0 < a < b)$$

Proof. Let f be the integrand on the left hand side. Notice the poles of f are all x satisfying $e^{bx} = -1$, which are precisely $\{x \mid x = i\frac{\pi(2k+1)}{b}, k \in \mathbb{Z}\}$. Now choose a rectangular contour which is the union of four sides, γ_i for $i = 1, 2, 3, 4$ corresponding to the bottom, right, top, and left sides of the contour which intersects the imaginary axis at $(2\pi i)/b$ and the real axis at $-R$ and R . First take the integral of

$$\frac{e^{az}}{e^{bz} + 1} dz$$

on γ_3 . Letting $z = x + i\frac{2\pi}{b}$ implies $dz = dx$ and so we have

$$\int_{\gamma_3} \frac{e^{ax}}{e^{bx} + 1} dx = - \int_{-R}^R \frac{e^{a(x+i\frac{2\pi}{b})}}{e^{b(x+i\frac{2\pi}{b})} + 1} dx = -e^{(2\pi ia)/b} \int_{-R}^R \frac{e^{ax}}{e^{bx} + 1} dx.$$

For γ_2 , notice $z = R + iy$ where $y \in [0, \frac{2\pi}{b}]$, so since yb is real, we have $|e^{bz}| = |e^{Rb}e^{iyb}| = e^{Rb}$. Notice we have an upper bound given by

$$\left| \frac{e^{az}}{e^{bz} + 1} \right| \leq \frac{e^{Ra}}{e^{Rb} - 1} = \frac{1}{e^{R(b-a)} - e^{-Ra}},$$

where we have divided all terms by e^{Ra} . Then for our ML bound we have

$$\left| \int_{\gamma_2} \frac{e^{az}}{e^{bz} + 1} dz \right| \leq \frac{2\pi}{b} \frac{1}{e^{R(b-a)} - e^{-Ra}},$$

which converges to 0 as $R \rightarrow \infty$ because $a < b$. Similarly one can show the integral on γ_4 converges to 0. Returning to γ_1 , we have a single pole that is located inside the contour when $k = 0$. Using the Residue theorem notice

$$\text{Res}(f, i\frac{\pi}{b}) = \frac{e^{az}}{be^{bz}} = -\frac{e^{(a/b)\pi i}}{b}.$$

Thus our desired integral has value

$$-2\pi i \left(\frac{e^{(a/b)\pi i}}{b} \right),$$

which implies as $R \rightarrow \infty$, we have

$$\int_{-R}^R \frac{e^{ax}}{e^{bx} + 1} dx = -\frac{2\pi i(e^{(a/b)\pi i})}{b - be^{(a/b)2\pi i}} = \frac{2\pi i}{b(e^{(a/b)\pi i} - e^{-(a/b)\pi i})}.$$

Dividing all terms by $2i$ now gives us the desired result, since we know

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

where $x = (a/b)\pi$ in this problem. □

1.3. **Q3.** Evaluate the improper integral

$$\int_0^\infty \frac{\sqrt{x}}{x^3 + 1} dx = \frac{\pi}{3}$$

Proof. We will use the substitution $x = e^t$, which implies $dx = e^t dt$ and also $\sqrt{x} = e^{t/2}$. Changing the limits of integration, the left hand side is equivalent to

$$\int_{-\infty}^\infty \frac{e^{t/2}}{e^{3t+1}} e^t dt = \int_{-\infty}^\infty \frac{e^{3x}}{e^{(3x)/2}} dx.$$

We know according to the previous question

$$\int_{-\infty}^\infty \frac{e^{ax}}{e^{bx} + 1} dx = \frac{\pi}{b \sin(\frac{\pi a}{b})}.$$

Using this identity with $a = 3/2$ and $b = 3$ (which satisfies $0 < a < b$), we have

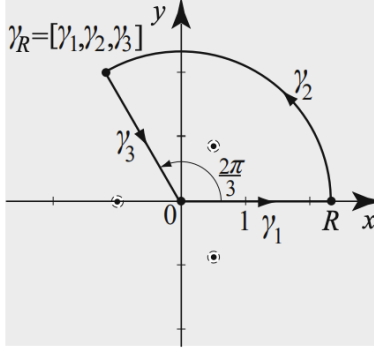
$$\int_{-\infty}^\infty \frac{e^{(3x)/2}}{e^{3x} + 1} = \frac{\pi}{3 \sin(\pi/2)} = \frac{\pi}{3}.$$

□

1.4. **Q4.** Use the contour γ_R in the figure below to evaluate

$$\int_0^\infty \frac{1}{x^3 + 1} dx.$$

[Hint: $I_{\gamma_R} = I_1 + I_2 + I_3$; $I_3 = -e^{i(\frac{2\pi}{3})}I_1$; and $I_2 \rightarrow 0$ as $R \rightarrow \infty$.]



Proof. We use the hint which states $I_{\gamma_R} = I_1 + I_2 + I_3$; $I_3 = -e^{i(\frac{2\pi}{3})}I_1$; and $I_2 \rightarrow 0$ as $R \rightarrow \infty$. To find the poles of the integrand, notice the equation $x^3 + 1 = 0$ has three solutions but only $x_1 = \cos(\pi/3) + i\sin(\pi/3)$ lies inside the contour. Now if we let $x_2 = \cos(2\pi/3) + i\sin(2\pi/3)$ and $x_3 = \cos(\pi) + i\sin(\pi) = -1$ are the other roots, then by the Residue theorem we know

$$\text{Res}(f, x_1) = \lim_{x \rightarrow x_1} \frac{1}{(x - x_2)(x - x_3)} = \frac{1}{(\frac{3}{2} + i\frac{\sqrt{3}}{2})(i\sqrt{3})} = I_{\gamma_R}.$$

Recall $I_{\gamma_R} = I_1 + I_2 + I_3$, but notice for I_2 , with $M = \frac{1}{R^3+1}$ and $L = \frac{2\pi R}{3}$, we have as $R \rightarrow 0$ the integral $\int \frac{1}{x^3+1} dx \leq \frac{1}{R^3+1} \frac{2\pi R}{3}$ converging to 0. Thus $I_2 \rightarrow 0$ as given by the hint. By the hint, notice

$$\frac{2\pi i}{(\frac{3}{2} + i\frac{\sqrt{3}}{2})(i\sqrt{3})} = \frac{2\pi}{(\frac{3}{2} + i\frac{\sqrt{3}}{2})(\sqrt{3})} = I_1(1 - e^{i(\frac{2\pi}{3})}).$$

Euler's formula implies

$$I_1(1 - e^{i(\frac{2\pi}{3})}) = I_1(1 - \cos(2\pi/3) - i\sin(2\pi/3))$$

so solving for I_1 yields

$$I_1 = \frac{2\pi}{(3/4 + 9/4)} = \frac{2\pi\sqrt{3}}{3\sqrt{3}\sqrt{3}} = \frac{2\pi\sqrt{3}}{9}$$

as desired. We conclude

$$\int_0^\infty \frac{1}{x^3 + 1} dx = \frac{2\pi\sqrt{3}}{9}.$$

□

1.5. **Q5.** Evaluate the improper integral

$$\int_{-\infty}^{\infty} \frac{\cos x}{(x^2 + 1)^2} dx = \frac{\pi}{e}.$$

Make sure to outline the steps in your solution. Carry out the details even if the integral follows from previously derived formulas. Here $a, b \in \mathbb{R}$ and $c > 0$.

Proof. To see if the given integral is convergent, let f be the integrand and notice

$$|f| \leq \frac{1}{(x^2 + 1)^2},$$

which is convergent. Notice

$$\int_{-\infty}^{\infty} \frac{\cos x}{(x^2 + 1)^2} dx = \int_{-\infty}^{\infty} \operatorname{Re} \left(\frac{e^{ix}}{(x^2 + 1)^2} \right) dx.$$

Consider the contour of integration γ_R defined by the union of γ_1 , which is the positively oriented interval $[-R, R]$ and γ_2 , which is the positively-oriented upper semi-circle that meets γ_1 at R and $-R$. Hence our desired integral

$$\int_{\gamma_R} \frac{e^{iz}}{(z^2 + 1)^2} dz$$

is the sum of the two integrals

$$\int_{\gamma_1} \frac{e^{ix}}{(x^2 + 1)^2} dx, \quad \text{and} \quad \int_{\gamma_2} \frac{e^{iz}}{(z^2 + 1)^2} dz.$$

Notice our integral is of the form

$$\lim_{R \rightarrow \infty} \int_{\sigma_R} e^{isz} \frac{p(z)}{q(z)} dz,$$

where $s > 0$ and the degree of $q = (z^2 + 1)^2$ is greater than the degree of $p = 1$. Therefore, Cor 5.4.4 (Asmar) implies that as $R \rightarrow \infty$, we have

$$\int_{\gamma_2} \frac{e^{iz}}{(z^2 + 1)^2} dz \rightarrow 0.$$

Thus

$$\int_{\gamma_R} \frac{e^{iz}}{(z^2 + 1)^2} dz = \int_{\gamma_1} \frac{e^{ix}}{(x^2 + 1)^2} dx.$$

The integrand has a pole of order 2 at $x = i$ and also a pole of order 2 at $x = -i$. But the latter is not located inside the contour, so we only find the residue of f at i . We have

$$\begin{aligned} \operatorname{Res} \left(\frac{e^{iz}}{(z^2 + 1)^2}, i \right) &= \lim_{z \rightarrow i} \frac{d}{dz} \left(\frac{e^{iz}}{(z + i)^2} \right) = \lim_{z \rightarrow i} \left(\frac{e^{iz} i (z + i)^2 - 2e^{iz} (z + i)}{(z + i)^4} \right) \\ &= \frac{i(-4 - 4)}{16e} = \left(\frac{-i}{2e} \right). \end{aligned}$$

But the residue theorem implies

$$(-2\pi i) \left(\frac{-i}{2e} \right) = \frac{\pi}{e} = \int_{\gamma_R} \frac{e^{iz}}{(z^2 + 1)^2} dz = \int_{\gamma_1} \frac{e^{ix}}{(x^2 + 1)^2} dx,$$

as $R \rightarrow \infty$. Since π/e has no imaginary part, we know this is equivalent to the integral

$$\int_{-\infty}^{\infty} \frac{\cos x}{(x^2 + 1)^2} dx.$$

□